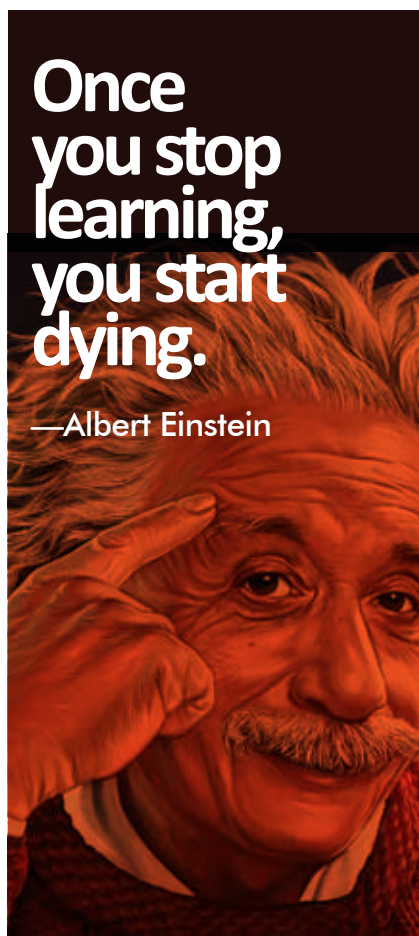




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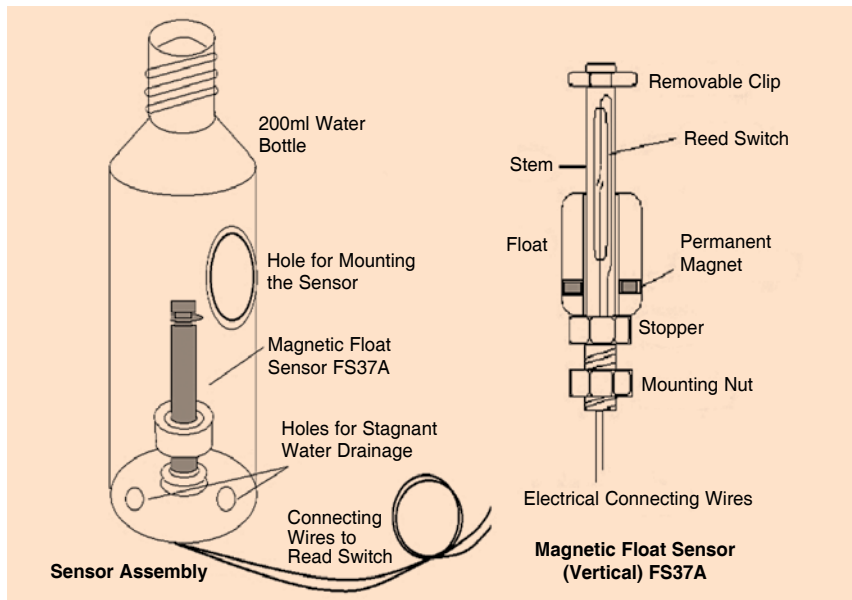


Fig. 3: Float sensor FS37A assembly

S-R LATCH TRUTH TABLE

S	R	Output (Pin 3)	Comment	T1 Collector Voltage	Relay/Pump Status
1	1	No change	Memory effect	No change	No change
	1	1	SET	0V (approx)	Turned off
1		0	RESET	+5V (approx)	Turned on
0	0		Not available		

Circuit and working

Fig. 5 shows the circuit diagram for controlling the relay. Here, a 555 timer (IC2) is configured as a bistable multivibrator (S-R latch). Trigger pin 2 of IC2 serves as the set (S) input, while reset pin 4 functions as the reset (R) input. Transistor T1 acts as a NOT gate, inverting the logic level of the output (pin 3) of the S-R latch. The truth table of the S-R latch is shown in the table.

The power supply circuits are self-explanatory. The +12V supply is obtained as an unregulated rectified output from step-down transformer X1 (a 230V to 12V-0-12V, 600mA centre-tapped transformer). Rectification is carried out by two 1N4007 diodes (D1 and D2), with capacitor C1 serving as the input filter. To derive a regulated +5V supply from this unregulated +12V supply, a three-terminal voltage



Fig. 4: Sensor assembly attached to the overflow pipe